

PDGFR α signaling regulates Srsf3 transcript binding to affect PI3K signaling and endosomal trafficking

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Abstract

Signaling through the platelet-derived growth factor receptor alpha (PDGFR α) plays a critical role in craniofacial development, as mutations in *PDGFRA* are associated with cleft lip/palate in humans and *Pdgfra* mutant mouse models display varying degrees of facial clefting. Phosphatidylinositol 3-kinase (PI3K)/Akt is the primary effector of PDGFR α signaling during skeletal development in the mouse. We previously demonstrated that Akt phosphorylates the RNA-binding protein serine/arginine-rich splicing factor 3 (Srsf3) downstream of PI3K-mediated PDGFR α signaling in mouse embryonic palatal mesenchyme (MEPM) cells, leading to its nuclear translocation. We further showed that ablation of *Srsf3* in the murine neural crest lineage results in severe midline facial clefting, due to defects in proliferation and survival of cranial neural crest cells, and widespread alternative RNA splicing (AS) changes. Here, we sought to determine the molecular mechanisms by which Srsf3 activity is regulated downstream of PDGFR α signaling to control AS of transcripts necessary for craniofacial development. We demonstrated via enhanced UV-crosslinking and immunoprecipitation (eCLIP) of MEPM cells that PDGF-AA stimulation leads to preferential binding of Srsf3 to exons and loss of binding to canonical Srsf3 CA-rich motifs. Through the analysis of complementary RNA-seq data, we showed that Srsf3 activity results in the preferential inclusion of exons with increased GC content and lower intron to exon length ratio. Moreover, we found that the subset of transcripts that are bound by Srsf3 and undergo AS upon PDGFR α signaling commonly encode regulators of PI3K signaling and early endosomal trafficking. Functional validation studies further confirmed that Srsf3 activity downstream of PDGFR α signaling leads to retention of the receptor in early endosomes and increases in downstream PI3K-mediated Akt signaling. Taken together, our findings reveal that growth factor-mediated phosphorylation of an RNA-binding protein underlies gene expression regulation necessary for mammalian craniofacial development.

eLife assessment

This **fundamental** work provides new mechanistic insight into the regulation of PDGF signaling through splicing controls. The evidence is **compelling** to demonstrate the involvement of Srsf3, an RNA-binding protein, in this new mechanism. The work will be of broad interest to developmental biologists in general and molecular biologists/biochemists in the field of growth factor signaling and RNA splicing.

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Introduction

Craniofacial development is a complex morphogenetic process that requires a precise interplay of multiple cell and tissue types to generate the frontonasal skeleton. Disruption of this process can result in some of the most common birth defects in humans, such as cleft lip and palate (Mai et al., 2019). Signaling through the platelet derived growth factor receptor alpha (PDGFR α) receptor tyrosine kinase (RTK) is essential for human craniofacial development.

Heterozygous missense mutations in the coding region of *PDGFRA* that alter amino acids within the extracellular, transmembrane or cytoplasmic domains of the receptor, in addition to single base-pair substitutions in the 3' untranslated region (3' UTR), are associated with nonsyndromic cleft palate (Rattanasopha et al., 2012). Further, single-nucleotide polymorphisms that repress transcriptional activity of the promoter upstream of *PDGFC*, which encodes one of two PDGFR α ligands, are associated with cleft lip and palate (Choi et al., 2009). This role of PDGFR α signaling in craniofacial development is conserved in mice, as *Pdgfra* mutant mouse models display a variety of defects that range from cleft palate to complete facial clefting (Klinghoffer et al., 2002; Soriano, 1997; Tallquist & Soriano, 2003; Fantauzzo & Soriano, 2014; He & Soriano, 2013). These phenotypes are recapitulated in embryos lacking both *Pdgfa* and *Pdgfc* (Ding et al., 2004). Phosphatidylinositol 3-kinase (PI3K) is the primary effector of PDGFR α signaling during skeletal development in the mouse (Klinghoffer et al., 2002). Following activation, PI3K increases phosphatidylinositol-3,4,5-trisphosphate (PIP $_3$) levels at the cell membrane, leading to the recruitment and subsequent phosphorylation of the serine/threonine kinase Akt. Akt subsequently dissociates from the membrane to phosphorylate an array of target proteins that are involved in wide-ranging cellular processes (Manning & Cantley, 2007). We previously identified proteins phosphorylated by Akt downstream of PI3K-mediated PDGFR α signaling in primary mouse embryonic mesenchyme (MEPM) cells (Fantauzzo & Soriano, 2014). Gene ontology analysis revealed that 25% of the 56 proteins were involved in RNA processing, with a particular enrichment for RNA splicing (Fantauzzo & Soriano, 2014).

Alternative RNA splicing (AS) is a process by which different combinations of exons from the same gene are incorporated into mature RNA transcripts, thereby contributing to gene expression regulation and enhancing the diversity of protein isoforms (Licatalosi & Darnell, 2010). AS occurs in approximately 95% of multi-exon human genes, frequently in a tissue-specific manner (Pan et al., 2008; Wang et al., 2008). Dysregulation of AS causes a number of diseases, due to mutations in precursor RNA sequences, mutations in core components of the spliceosome complex and/or mutations in auxiliary RNA-binding proteins (RBPs) (Scotti & Swanson, 2016). These *trans*-acting auxiliary RBPs bind to specific sequence and/or structural motifs in a target RNA via one or more RNA-binding domains to promote or inhibit exon inclusion (Fu & Ares, 2014; Licatalosi & Darnell, 2010). The phenotypes resulting from global and/or tissue-specific knockout of multiple RBPs have established that RBP-mediated AS is an essential process during mouse

craniofacial development (Bebbee et al., 2015 [↗](#); Cibi et al., 2019 [↗](#); Dennison et al., 2021 [↗](#); Forman et al., 2021 [↗](#); Lee et al., 2020 [↗](#)). We previously demonstrated that ablation of *Srsf3* in the murine neural crest lineage results in severe midline facial clefting and facial bone hypoplasia, due to defects in proliferation and survival of cranial neural crest cells, and widespread AS changes (Dennison et al., 2021 [↗](#)).

Srsf3 belongs to the serine/arginine-rich (SR) protein family of splicing factors that generally promote exon inclusion by binding to exonic and intronic splicing enhancers and by recruiting spliceosome components to the 5' and 3' splice sites (Fu & Ares, 2014 [↗](#); Licatalosi & Darnell, 2010 [↗](#)). *Srsf3* specifically was shown to bind pyrimidine-rich motifs, with a preference for exonic regions (Akerman et al., 2009 [↗](#); Änkö et al., 2012 [↗](#)). *Srsf3* is phosphorylated downstream of PDGF and EGF stimulation and all-*trans* retinoic acid treatment, and by the kinases Akt and SRPK2 (Bavelloni et al., 2014 [↗](#); Dennison et al., 2021 [↗](#); Fantauzzo & Soriano, 2014 [↗](#); Long et al., 2019 [↗](#); Zhou et al., 2012 [↗](#)). Phosphorylation of the Akt consensus sites within the C-terminal arginine/serine-rich (RS) domain of *Srsf3* drives its translocation to the nucleus, where AS takes place (Bavelloni et al., 2014 [↗](#); Dennison et al., 2021 [↗](#); Long et al., 2019 [↗](#)). Moreover, phosphorylation of the RS domain can alter the ability of SR proteins to interact with other proteins, such as the U1 small nuclear ribonucleoprotein (snRNP) component of the spliceosome and additional RBPs, and affect the ability of SR proteins to bind RNA (Huang et al., 2004 [↗](#); Shen & Green, 2006 [↗](#); Shin et al., 2004 [↗](#); Xiao & Manley, 1997 [↗](#)). However, the molecular mechanisms by which *Srsf3* activity is regulated downstream of specific signaling inputs in a context-specific manner to regulate RNA binding and/or sequence specificity remain undetermined.

Here, we identified changes in *Srsf3*-dependent AS and *Srsf3* transcript binding in the absence or presence of PDGF-AA ligand in MEPM cells. RNA-sequencing (RNA-seq) analysis revealed that *Srsf3* activity and PDGFR α signaling have more pronounced effects on AS than gene expression, as well as a significant dependence in regulating AS. Using enhanced UV-crosslinking and immunoprecipitation (eCLIP), we found a shift from intronic to exonic *Srsf3* binding and loss of CA-rich sites upon PDGF-AA stimulation. Further, we demonstrated that the subset of transcripts that are bound by *Srsf3* and undergo AS upon PDGFR α signaling commonly encode regulators of PI3K signaling and early endosomal trafficking, ultimately serving as a feedback mechanism to affect trafficking of the receptors. Combined, our findings provide significant insight into the mechanisms underlying RBP-mediated gene expression regulation in response to growth factor stimulation within the embryonic mesenchyme.

Results

PDGFR α signaling for one hour minimally affects gene expression

To determine the *Srsf3*-dependent changes in gene expression and AS downstream of PDGFR α signaling, we stably integrated a scramble shRNA (scramble) or an shRNA targeting the 3' UTR of *Srsf3* (sh*Srsf3*) into immortalized MEPM (iMEPM) cells via lentiviral transduction (Fig. 1A [↗](#)). Western blotting revealed a 66% decrease in *Srsf3* protein levels in the sh*Srsf3* cell line (Fig. 1B [↗](#)). We previously demonstrated that phosphorylated *Srsf3* levels peaked in the nucleus of iMEPM cells following 1 hour of PDGF-AA ligand treatment (Dennison et al., 2021 [↗](#)). As such, cells were left unstimulated (-PDGF-AA) or treated with 10 ng/mL PDGF-AA for 1 hour (+PDGF-AA). RNA was isolated and sequenced from three biological replicates across each of the four conditions (Fig. 1A [↗](#); Table S1). We first examined *Srsf3*-dependent differentially-expressed (DE) genes by comparing scramble versus sh*Srsf3* samples across the same PDGF-AA stimulation condition (-PDGF-AA or +PDGF-AA). We detected 827 DE genes in the absence of ligand treatment and 802 DE genes upon ligand stimulation (Fig. 1C [↗](#), Table S2). There was high overlap (1,042 out of 1,629; 64.0%) of DE genes across ligand treatment conditions (Fig. 1C,D [↗](#); Fig. S1A). Of these 521 shared DE genes, 514 (98.7%) had the same directionality, including 273 (52%) with shared increases in

expression in the shSrsf3 samples and 241 (46%) with shared decreases in expression in the shSrsf3 samples (**Fig. 1C,D**). These findings demonstrate that expression of a set of genes (521) depends on Srsf3 activity independent of PDGFR α signaling, while a similarly sized set of genes (587) is differentially expressed in response to both Srsf3 activity and PDGFR α signaling (**Fig. 1C,D**). Gene ontology (GO) analysis of the Srsf3-dependent DE genes revealed that the most significant terms for biological process commonly involved regulation of osteoblast differentiation, calcium-dependent cell-cell adhesion, regulation of cell migration and canonical Wnt signaling, while only a handful of significant terms for molecular function were detected in unstimulated cells, relating to cation channel activity and phosphatase activity (Fig. S2A,B).

We next examined PDGF-AA-dependent DE genes by comparing -PDGF-AA versus +PDGF-AA across the same Srsf3 condition (scramble or shSrsf3). We detected only 37 DE genes in scramble cells and 14 DE genes in shSrsf3 cells (**Fig. 1E**, Table S2). There was low overlap (8 out of 51; 15.7%) of DE genes across Srsf3 conditions (**Fig. 1E,F**; Fig. S1B). Of these 4 shared DE genes, 3 (75%) had shared increases in expression upon PDGF-AA stimulation (**Fig. 1E,F**). These findings demonstrate that, unlike Srsf3 activity, PDGFR α signaling minimally affects gene expression at one hour of ligand stimulation, consistent with our previous findings in mouse embryonic facial mesenchyme (Dennison et al., 2021). Further, these results show that expression of a set of genes (4) depends on PDGFR α signaling independent of Srsf3 activity, while a larger set of genes (43) is differentially expressed in response to both PDGFR α signaling and Srsf3 activity (**Fig. 1E,F**). GO analysis of the PDGF-AA-dependent DE genes revealed significant terms for biological process in the scramble cells relating to cell migration, response to growth factor stimulation and regulation of transcription (Fig. S2A). The most significant terms for molecular function commonly involved DNA binding (Fig. S2B).

PDGFR α signaling for one hour has a more pronounced effect on alternative RNA splicing

We previously demonstrated that AS is an important mechanism of gene expression regulation downstream of PI3K/Akt-mediated PDGFR α signaling in the murine mid-gestation palatal shelves (Dennison et al., 2021). Accordingly, we next assessed AS in our same RNA-seq dataset. In examining Srsf3-dependent alternatively-spliced transcripts, we detected 1,354 differential AS events in the absence of ligand treatment and 1,071 differential AS events upon ligand stimulation (**Fig. 2A**). When filtered to include events detected in at least 10 reads in either condition, we obtained a list of 1,113 differential AS events in the absence of ligand treatment and 795 differential AS events upon ligand stimulation (**Fig. 2B**, Tables S3 and S4). There was low overlap (406 out of 1,908; 21.3%) of alternatively-spliced transcripts across ligand treatment conditions. Of these 203 shared alternatively-spliced transcripts, 100% had the same directionality, including 81 (40%) with shared negative changes in percent spliced in (PSI) (exon included more often in shSrsf3 samples) and 122 (60%) with shared positive changes in PSI (exon skipped more often in shSrsf3 samples) (**Fig. 2A,B**). We confirmed the differential AS of two transcripts, *Arhgap12* and *Cep55*, between scramble and shSrsf3 samples by qPCR using primers within constitutively-expressed exons flanking the alternatively-spliced exon (Fig. S3A,B). These findings demonstrate that AS of a set of transcripts (203) depends on Srsf3 activity independent of PDGFR α signaling, while a much larger set of transcripts (1,502) is alternatively spliced in response to both Srsf3 activity and PDGFR α signaling (**Fig. 2A,B**). GO analysis of these Srsf3-dependent alternatively-spliced transcripts revealed that the most significant terms for biological process and molecular function commonly involved regulation of RNA splicing, and RNA binding and cadherin binding, respectively (Fig. S4A,B).

In examining PDGF-AA-dependent alternatively-spliced transcripts, we detected 595 differential AS events in scramble cells and 398 differential AS events in shSrsf3 cells (**Fig. 2C**). When filtered to include events detected in at least 10 reads in either condition, we obtained a list of 375 differential AS events in scramble cells and 256 differential AS events in shSrsf3 cells (**Fig. 2D**,

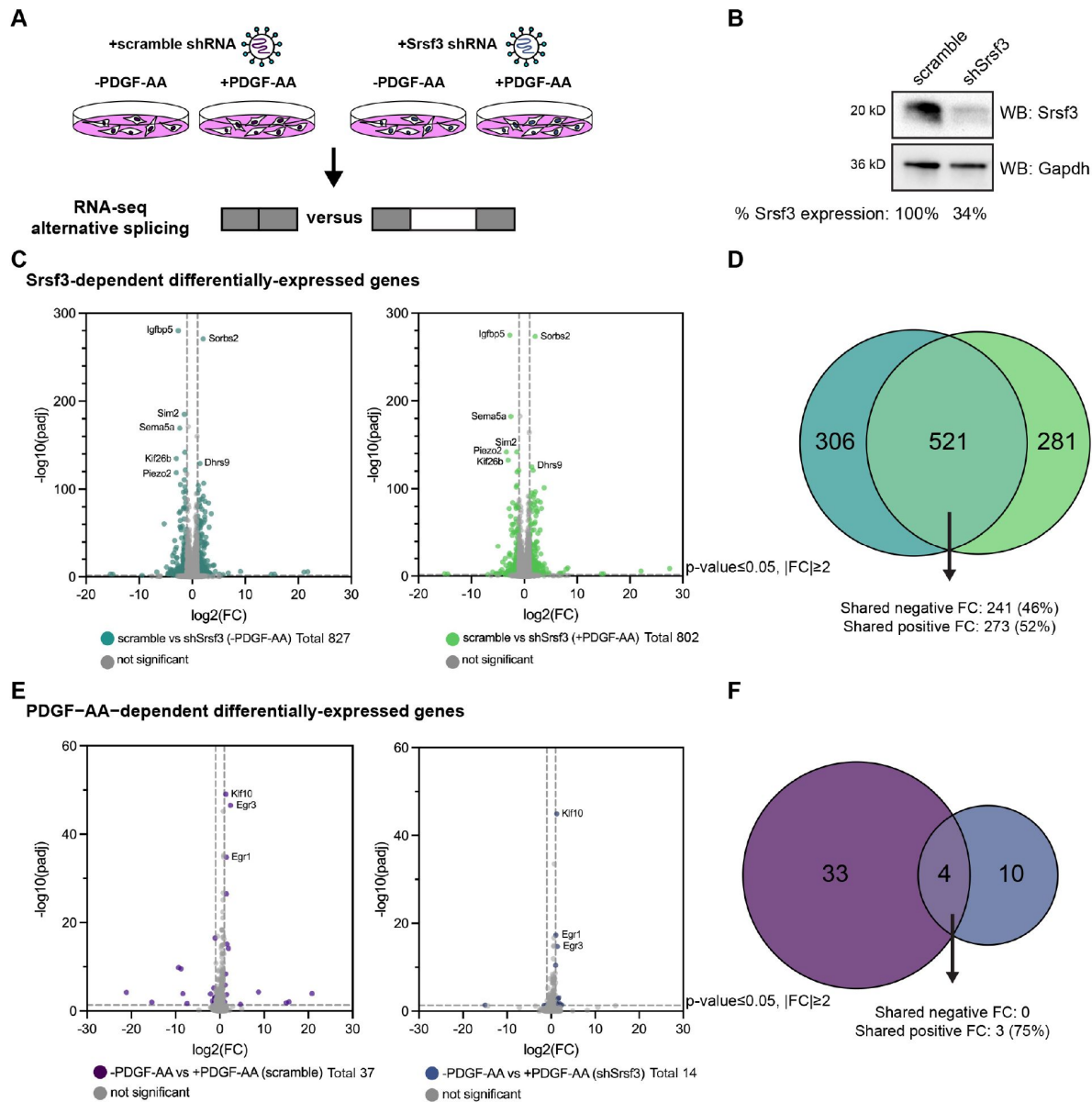


Figure 1

PDGFR α signaling for one hour minimally affects gene expression.

(A) Schematic of RNA-seq experimental design. iMEPM cells were transduced to stably express a scramble shRNA (scramble) or shRNA targeting the 3' UTR of *Srsf3* (shSrsf3). iMEPM cells expressing either scramble or shSrsf3 were left unstimulated or stimulated with 10 ng/mL PDGF-AA for 1 hour and RNA was isolated for RNA-seq analysis. (B) Western blot (WB) analysis of whole-cell lysates from scramble and shSrsf3 cell lines with anti-Srsf3 and anti-Gapdh antibodies. The percentage of Srsf3 expression normalized to Gapdh expression is indicated below. (C) Volcano plots depicting differentially-expressed genes in scramble versus shSrsf3 cell lines in the absence (left) or presence (right) of PDGF-AA stimulation. Log₂(fold change) (FC) values represent log₂(shSrsf3 normalized counts/scramble normalized counts). Significant changes in gene-level expression are reported for genes with adjusted *P* (padj) < 0.05 and fold change |FC| ≥ 2. (D) Venn diagram of significant genes from C. (E) Volcano plots depicting differentially-expressed genes in the absence versus presence of PDGF-AA ligand in scramble (left) or shSrsf3 (right) cell lines. Log₂(FC) values represent log₂(+PDGF-AA normalized counts/-PDGF-AA normalized counts). (F) Venn diagram of significant genes from E.

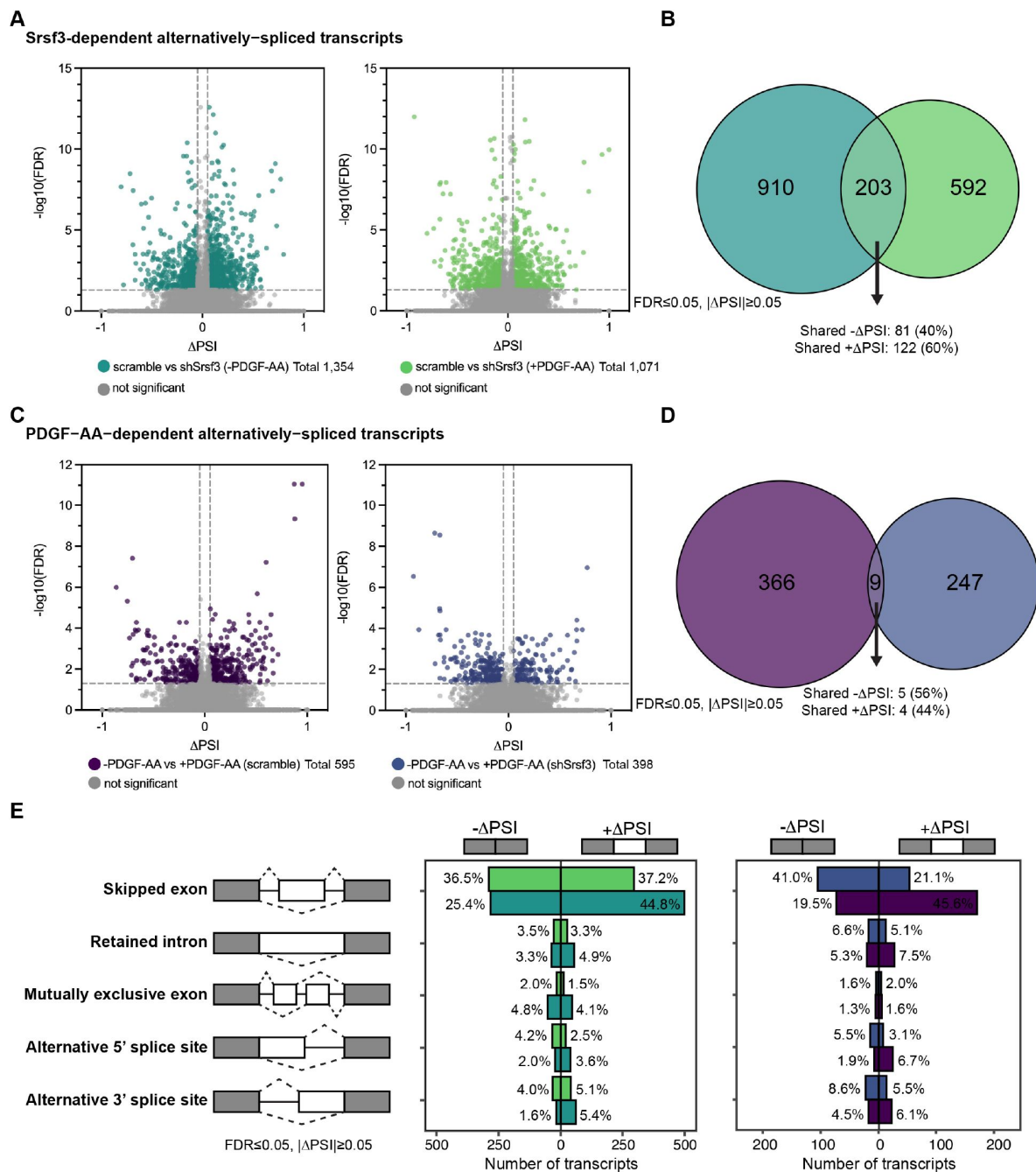


Figure 2

PDGFR α signaling for one hour has a more pronounced effect on alternative RNA splicing.

(A) Volcano plots depicting alternatively-spliced transcripts in scramble versus shSrsf3 cell lines in the absence (left) or presence (right) of PDGF-AA stimulation. Difference in percent spliced in (Δ PSI) values represent scramble PSI – shSrsf3 PSI. Significant changes in alternative RNA splicing are reported for events with a false discovery rate (FDR) ≤ 0.05 and a difference in percent spliced in ($|\Delta$ PSI|) ≥ 0.05 . (B) Venn diagram of significant transcripts from A, filtered to include events detected in at least 10 reads in either condition. (C) Volcano plots depicting alternatively-spliced transcripts in the absence versus presence of PDGF-AA ligand in scramble (left) or shSrsf3 (right) cell lines. Difference in percent spliced in (Δ PSI) values represent -PDGF-AA PSI – +PDGF-AA PSI. (D) Venn diagram of significant transcripts from C, filtered to include events detected in at least 10 reads in either condition. (E) Bar graph depicting alternative RNA splicing events in scramble versus shSrsf3 cell lines in the absence or presence of PDGF-AA stimulation (left) or in the absence versus presence of PDGF-AA ligand in scramble or shSrsf3 cell lines (right).

Tables S5 and S6). There was extremely low overlap (18 out of 631; 2.85%) of alternatively-spliced transcripts across Srsf3 conditions (**Fig. 2C,D**). Of these 9 shared alternatively-spliced transcripts, 100% had the same directionality, including 5 (56%) with shared negative changes in PSI (exon included more often in +PDGF-AA samples) and 4 (44%) with shared positive changes in PSI (exon skipped more often in +PDGF-AA samples) (**Fig. 2C,D**). Taken together, these findings demonstrate that both Srsf3 activity and PDGFR α signaling have more pronounced effects on AS than gene expression. While more transcripts and genes are subject to these regulatory mechanisms upon loss of Srsf3 activity, the magnitude of events is more greatly skewed towards AS upon PDGF-AA stimulation. Further, these results show that AS of a set of transcripts (9) depends on PDGFR α signaling independent of Srsf3 activity, while a much larger set of transcripts (613) is alternatively spliced in response to both PDGFR α signaling and Srsf3 activity (**Fig. 2C,D**). When combined with the data above (**Fig. 2B**), this points to a profound dependence between PDGFR α signaling and Srsf3 in regulating AS of transcripts in the facial mesenchyme. GO analysis of these PDGF-AA-dependent alternatively-spliced transcripts demonstrated only a handful of significant terms for molecular function in the scramble cells, all relating to protein kinase activity (**Fig. S4B**).

The vast majority of Srsf3-dependent (70.2-73.7%) and PDGF-AA dependent (62.1-65.1%) AS events involved skipped exons, with minimal contribution from retained introns, mutually exclusive exons, alternative 5' splice sites, or alternative 3' splice sites (**Fig. 2E**). For the Srsf3-dependent skipped exon events, there were more transcripts with + Δ PSI (exon skipped more often in shSrsf3 samples) (44.8%) as opposed to - Δ PSI (exon included more often in shSrsf3 samples) (25.4%) in the absence of PDGF-AA stimulation (**Fig. 2E**), consistent with previous results that SR proteins tend to promote exon inclusion (Fu & Ares, 2014; Licatalosi & Darnell, 2010). However, PDGF-AA ligand treatment led to an increase in the percentage of transcripts with - Δ PSI (36.5%) (**Fig. 2E**), indicating that PDGFR α signaling promotes exon skipping in the presence of Srsf3. Among the PDGF-AA-dependent skipped exon events, there was a significant shift in transcripts with - Δ PSI (exon included more often in +PDGF-AA samples) in the absence (41.0%) versus presence (19.5%) of Srsf3, and a corresponding shift in transcripts with + Δ PSI (exon skipped more often in +PDGF-AA samples) (21.1% and 45.6%, respectively) (**Fig. 2E**), again suggesting that PDGF-AA stimulation causes increased exon skipping when Srsf3 is present.

Srsf3 exhibits differential transcript binding upon PDGFR α signaling

To determine direct binding targets of Srsf3 downstream of PDGFR α signaling, we conducted eCLIP (Van Nostrand et al., 2016, 2017) of iMEPM cells that were left unstimulated (-PDGF-AA) or treated with 10 ng/mL PDGF-AA for 1 hour (+PDGF-AA) as above (**Fig. 3A**; Table S7). Immunoprecipitation with a previously validated Srsf3 antibody (Dennison et al., 2021) successfully enriched for Srsf3 in UV-crosslinked cells (**Fig. 3B**). We detected 6,555 total eCLIP peaks in protein-coding genes in the -PDGF-AA samples and 8,584 total peaks in the +PDGF-AA samples (Table S8). Among the -PDGF-AA peaks, 3,727 (56.9%) were located in exons (CDS) and 1,690 (25.8%) were located within introns, with the rest binding within 5' UTRs (768, 11.7%) and 3' UTRs (367, 5.60%) (**Fig. 3C,D**; Table S9). We observed substantial shifts in Srsf3-bound regions upon PDGF-AA stimulation. Many more +PDGF-AA peaks were located within exons (7,139, 83.2%), and less were located within introns (765, 8.91%), 5' UTRs (389, 4.53%) and 3' UTRs (287, 3.34%) (**Fig. 3C,D**; Table S9). Given that SR proteins are crucial for exon definition and bind to exonic splicing enhancers to recruit and stabilize core splicing machinery (Fu & Maniatis, 1990; Krainer et al., 1991; Zahler et al., 1993), we investigated Srsf3 binding around 5' and 3' splice sites in response to PDGFR α signaling. Consistent with the results above and previous findings (Änkö et al., 2012), Srsf3 binding was enriched in exonic regions, as opposed to intronic regions, surrounding the splice sites (**Fig. 3E**). There was greater mean coverage of Srsf3 peaks in the +PDGF-AA condition versus the -PDGF-AA condition within 100 nucleotides upstream of the 5' splice site and within 100 nucleotides downstream of the 3' splice site (**Fig. 3E**). Additionally, we

detected decreased mean coverage in the +PDGF-AA condition within 25 nucleotides downstream of the 5' splice site boundary (**Fig. 3E**). Taken together, these data show that PDGFR α signaling leads to increased binding of Srsf3 to exons. Finally, we performed unbiased motif enrichment analysis of Srsf3 peaks in unstimulated and PDGF-AA-treated samples, revealing that the mostly highly enriched motifs in -PDGF-AA samples were CACACA and AAGAAG (**Fig. 3F**). Of note, these CA-rich motifs have previously been identified as canonical Srsf3 motifs in a CLIP study utilizing a stably integrated SRSF3-EGFP transgene (Änkö et al., 2012). However, in PDGF-AA-stimulated samples, the most highly enriched motifs were GAAGCG, GAAGAA, and AGAAGA (**Fig. 3G**), suggesting that PDGFR α signaling influences Srsf3 binding specificity.

Srsf3 and PDGFR α signaling are associated with differential GC content and length of alternatively-spliced exons

We next probed our RNA-seq dataset to determine whether specific transcript features were associated with Srsf3-dependent AS. When comparing significant AS events between scramble and shSrsf3 samples in the absence of PDGF-AA stimulation, we found that included exons had a significantly higher GC content (median value of 53.4%) than both skipped exons (50.0%) and exons that were not differentially alternatively spliced (51.2%) when Srsf3 is present (**Fig. 4A**, Table S10). Additionally, we observed that included exons had a significantly lower ratio of downstream intron to exon GC content (0.856) than both skipped exons (0.901) and exons that were not differentially alternatively spliced (0.888) in the presence of Srsf3 (**Fig. 4B**, Table S10). The same comparisons revealed that the ratio of upstream and downstream intron to exon length was significantly decreased in included exons (median values of 12.4 and 10.6, respectively) as compared to both skipped exons (19.5 and 20.0) and exons that were not differentially alternatively spliced (14.2 and 13.3) in the presence of Srsf3 (**Fig. 4C,D**, Table S10). Taken together, our data demonstrate that exons which are included in the presence of Srsf3 tend to have a higher GC content and lower intron to exon length ratio.

To determine whether PDGFR α signaling had an effect on the transcript features to which Srsf3 bound, we subsequently examined our eCLIP dataset. We found that the exon GC content was significantly increased in exons bound by Srsf3 in the absence of ligand treatment (median value of 57.9%) as compared to unbound exons (51.5%) (**Fig. 4E**, Table S10). However, exon GC content was similar between unbound exons and those bound by Srsf3 upon PDGF-AA stimulation (53.8%) (**Fig. 4E**, Table S10). These findings indicate that PDGFR α signaling mediates binding of Srsf3 to exons with a lower GC content.

Transcripts bound by Srsf3 that undergo alternative splicing upon PDGFR α signaling encode regulators of PI3K signaling

To determine which transcripts are directly bound by Srsf3 and subject to DE and/or AS, we cross-referenced the eCLIP and RNA-seq datasets. We collated transcripts with an Srsf3 eCLIP peak that were uniquely detected in the -PDGF-AA or +PDGF-AA samples (2,660 transcripts across 3,388 peaks) (Tables S8 and S11). Similarly, we gathered differentially-expressed genes (596) or differentially alternatively-spliced transcripts (985) uniquely found in one of the four treatment comparisons (**Fig. 5A**, Table S11). Only 32 (5.4%) of the DE genes were directly bound by Srsf3, while 233 (23.7%) of the alternatively-spliced transcripts had an Srsf3 eCLIP peak, with very little overlap (1 transcript) between all three categories (**Fig. 5A**).

We next correlated the eCLIP peaks with AS events across all four treatment comparisons by identifying transcripts in which Srsf3 bound within an alternatively-spliced exon or within 250 bp of the neighboring introns (Tables S12-S15). In agreement with our findings above for the entire eCLIP dataset, Srsf3 exhibited differential binding in exons and surrounding the 5' and 3' splice sites upon PDGF-AA stimulation (**Fig. S5A,B**). We performed an unbiased motif enrichment

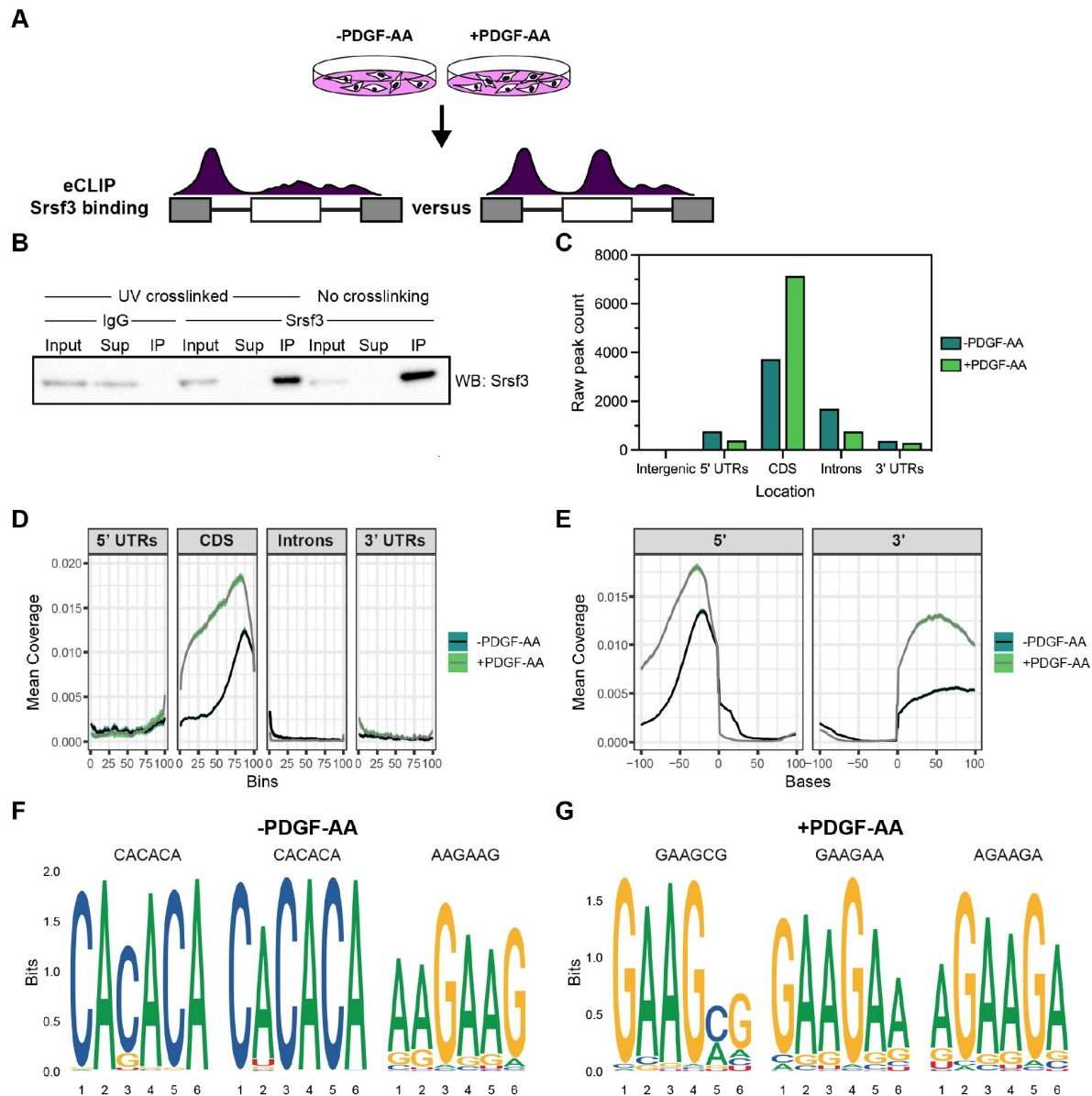


Figure 3

Srsf3 exhibits differential transcript binding upon PDGFR α signaling.

(A) Schematic of eCLIP experimental design. iMEPM cells were left unstimulated or stimulated with 10 ng/mL PDGF-AA for 1 hour and processed for eCLIP analysis. (B) Immunoprecipitation (IP) of Srsf3 from cells that were UV-crosslinked with IgG or an anti-Srsf3 antibody followed by western blotting (WB) of input, supernatant (Sup), and IP samples with an anti-Srsf3 antibody. (C) Mapping of eCLIP peaks to various transcript locations in the absence or presence of PDGF-AA stimulation. 5' UTR, 5' untranslated region; CDS, coding sequence; 3' UTR, 3' untranslated region. (D,E) Mean coverage of eCLIP peaks across various transcript locations (D) and surrounding the 5' and 3' splice sites (E) in the absence or presence of PDGF-AA stimulation. (F,G) Top three motifs enriched in eCLIP peaks in the absence (F) or presence (G) of PDGF-AA stimulation.

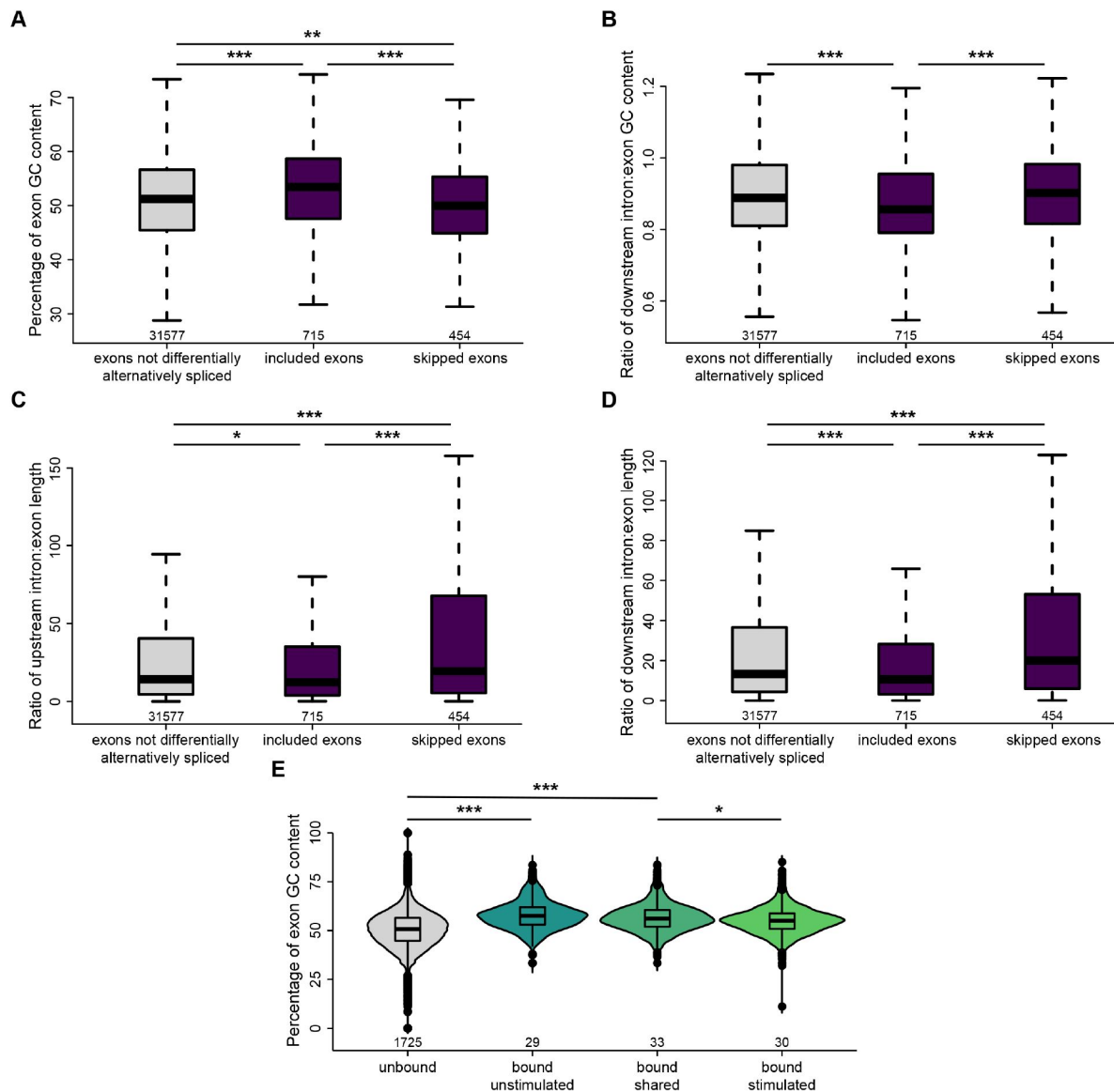


Figure 4

Srsf3 and PDGFR α signaling are associated with differential GC content and length of alternatively-spliced exons.

(A) Box and whisker plot depicting the percentage of exon GC content in exons that are not differentially alternatively spliced, and exons that are included or skipped when Srsf3 is present from the rMATS analysis. (B) Box and whisker plot depicting the ratio of downstream intron to exon GC content in exons that are not differentially alternatively spliced, and exons that are included or skipped when Srsf3 is present from the rMATS analysis. (C,D) Box and whisker plots depicting the ratio of upstream intron to exon length (C) and downstream intron to exon length (D) in exons that are not differentially alternatively spliced, and exons that are included or skipped when Srsf3 is present from the rMATS analysis. (E) Violin and box and whisker (inset) plots depicting the percentage of exon GC content in exons that are not bound by Srsf3, and exons that are bound in the absence and/or presence of PDGF-AA stimulation from the eCLIP analysis. *, $P < 0.05$; **, $P < 0.01$; ***, $P < 0.001$.

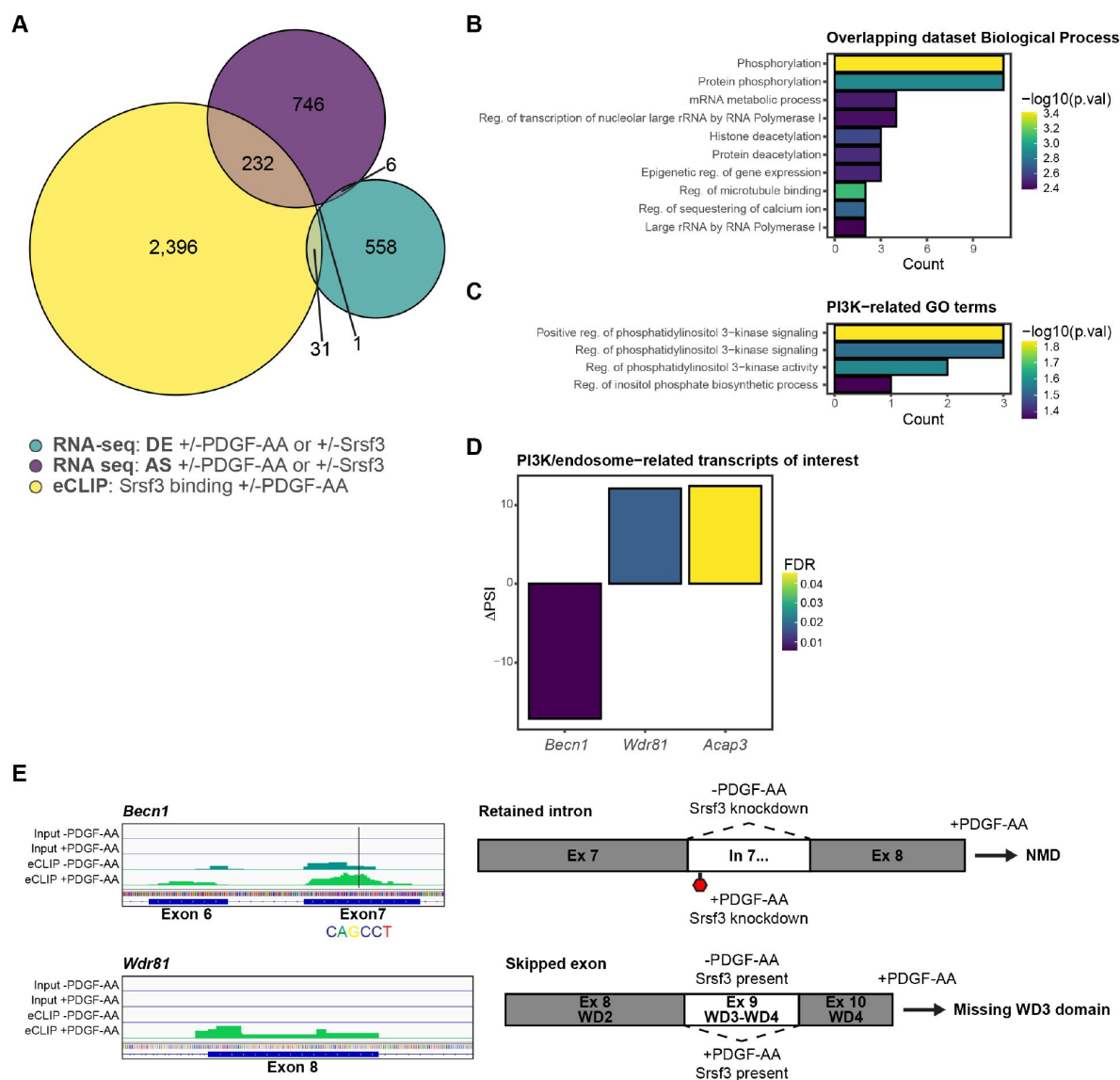


Figure 5

Transcripts bound by Srsf3 that undergo alternative splicing upon PDGFRα signaling encode regulators of PI3K signaling.

(A) Venn diagram of genes with differential expression (DE) or transcripts subject to alternative RNA splicing (AS) across the four treatment comparisons that overlap with transcripts with Srsf3 eCLIP peaks in the absence or presence of PDGF-AA stimulation. (B,C) Top ten (B) and PI3K-related (C) biological process gene ontology (GO) terms for transcripts from the overlapping datasets. p.val, *P*. (D) Difference in percent spliced in (ΔPSI) values for PI3K/endosome-related transcripts of interest. FDR, false detection rate. (E) Peak visualization for input and eCLIP samples in the absence or presence of PDGF-AA stimulation from Integrative Genomics Viewer (left) with location of motifs from Figure S5 indicated below for PI3K/endosome-related transcripts of interest. Predicted alternative splicing outcomes for PI3K/endosome-related transcripts of interest (right).

analysis of Srsf3 peaks within the overlapping datasets, again revealing different motifs between ligand treatment conditions and an enrichment of GA-rich motifs in the +PDGF-AA samples (Fig. S5C,D).

To determine whether transcripts that are differentially bound by Srsf3 and undergo differential AS downstream of PDGFR α signaling contribute to shared biological outputs, we conducted GO analysis of the 149 unique transcripts from the overlapping datasets. The most significant terms for biological process involved protein phosphorylation and deacetylation, and RNA metabolism (Fig. 5B). Given that PI3K-mediated PDGFR α signaling is critical for craniofacial development and regulates AS in this context (Dennison et al., 2021; Fantauzzo & Soriano, 2014; Klinghoffer et al., 2002), we turned our focus toward GO terms associated with this signaling pathway. We noted enrichment of PI3K-related GO terms (Fig. 5C), which encompassed the genes *Becn1*, *Wdr81* and *Acap3* (Fig. 5D). Related to their roles in PI3K signaling, each of these gene products has been shown to regulate membrane and/or endocytic trafficking. Phosphatidylinositol 3-phosphate (PI(3)P) is a critical component of early endosomes and is mainly generated by conversion of phosphatidylinositol (PI) by the class III PI3K complex, which includes Beclin-1 (encoded by *Becn1*) (Wallroth & Haucke, 2018). WDR81 and Beclin-1 have been shown to interact, resulting in decreased endosomal PI(3)P synthesis, PI(3)P turnover and early endosome conversion to late endosomes (Liu et al., 2016). Importantly, this role of WDR81 contributes to RTK degradation (Rapiteanu et al., 2016).

Within our data, Srsf3 binding was increased in *Becn1* exon 7 upon PDGF-AA stimulation, at an enriched motif within the overlapping datasets, and we observed a corresponding increase in retention of adjacent intron 7 (Fig. 5D,E). As *Becn1* intron 7 contains a premature termination codon (PTC), this event is predicted to result in nonsense-mediated decay (NMD) in the absence of Srsf3 (Fig. 5E). Srsf3 binding was also increased in *Wdr81* exon 8 in response to PDGF-AA treatment, and our analyses revealed a corresponding increase in excision of adjacent exon 9 (Fig. 5D,E). Because *Wdr81* exon 9 encodes two WD-repeat domains, which are generally believed to form a Δ propeller structure required for protein interactions (Li & Roberts, 2001), this AS event is predicted to result in a protein missing a functional domain (Fig. 5E). These splicing patterns predict increased levels of Beclin-1 and decreased levels of functional Wdr81 in the presence of Srsf3 and PDGF-AA stimulation, resulting in augmented early endosome formation.

Srsf3 regulates early endosome size and phosphorylation of Akt downstream of PDGFR α signaling

Given that Srsf3 differentially binds to transcripts that encode proteins involved in early endosomal trafficking downstream of PDGFR α signaling, we first examined the formation of Rab5-positive early endosomes (Zerial & McBride, 2001) in response to a time course of PDGF-AA ligand stimulation in scramble versus shSrsf3 cells. As expected, Rab5 puncta size increased from 0 min ($9.79 \times 10^{-4} \pm 1.13 \times 10^{-4}$ arbitrary units; mean $\pm$ s.e.m.) to 15 min of ligand stimulation ($2.01 \times 10^{-3} \pm 7.20 \times 10^{-4}$ arbitrary units) in scramble cells, and significantly so by 60 min ($2.58 \times 10^{-3} \pm 9.20 \times 10^{-4}$ arbitrary units) (Fig. 6A, C-C", E-E", G-G"). However, this increase was not observed in the absence of Srsf3 ($9.21 \times 10^{-4} \pm 1.61 \times 10^{-4}$ arbitrary units at 60 min) (Fig. 6A, D-D", F-F", H-H"), demonstrating that Srsf3-mediated PDGFR α signaling leads to enlarged early endosomes.

We next examined colocalization of PDGFR α with Rab5, as an estimate of receptor levels in the early endosome. Colocalization levels increased from 0 min (0.332 ± 0.0832 Pearson's correlation coefficient (PCC); mean $\pm$ s.e.m.) to 15 min (0.429 ± 0.108 PCC) of PDGF-AA treatment in scramble cells, then decreased to near baseline levels by 60 min (0.348 ± 0.0885 PCC) (Fig. 6B, C-C", E-E", G-G") as a subset of PDGFR α homodimers are likely trafficked to late endosomes (Rogers et al., 2022). Strikingly, shSrsf3 cells exhibited a significant decrease in colocalization between

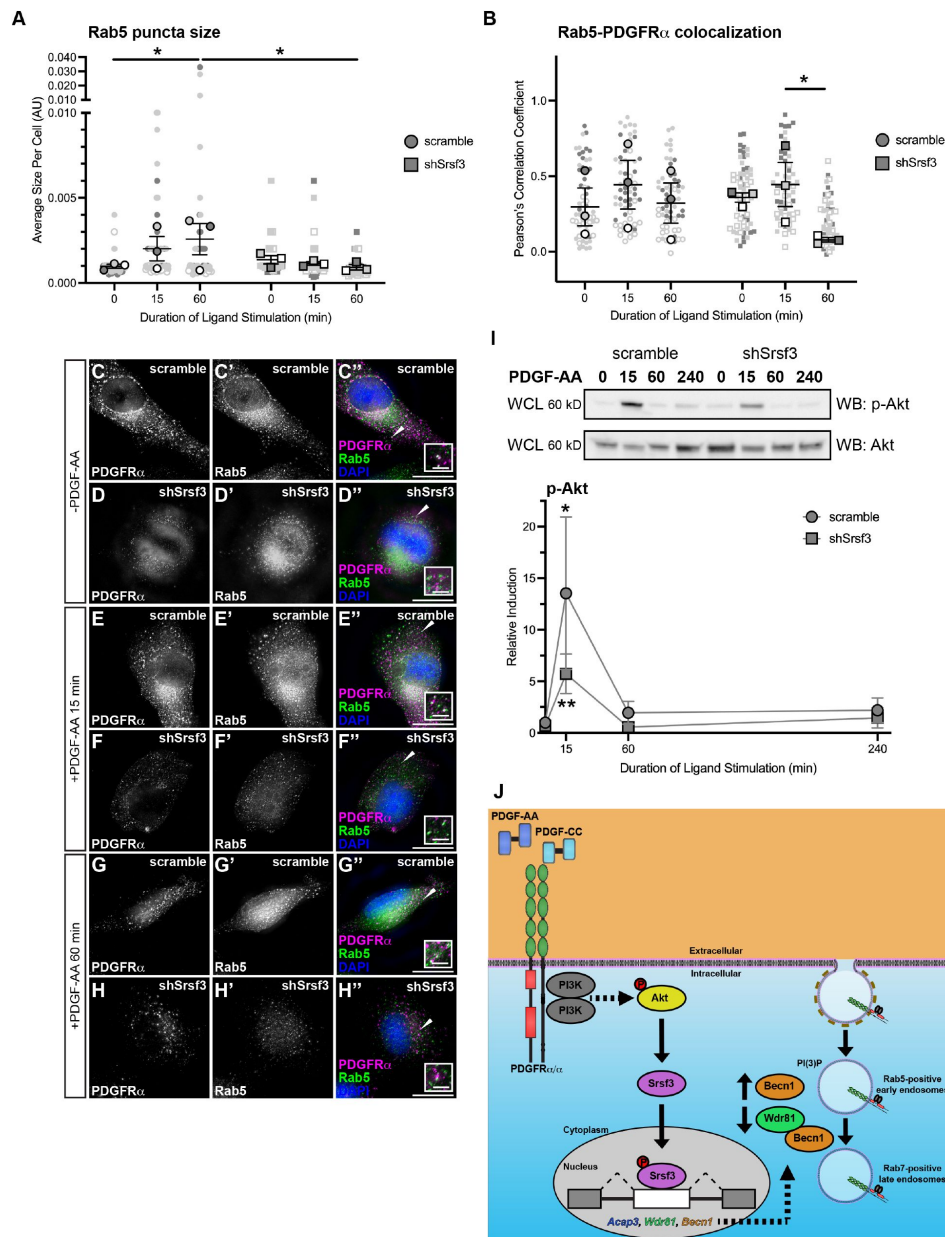


Figure 6

Srsf3 regulates early endosome size and phosphorylation of Akt downstream of PDGFR α signaling.

(A,B) Scatter dot plots depicting average size of Rab5 puncta per cell (A) and Pearson's correlation coefficient of signals from anti-Rab5 and anti-PDGFR α antibodies (B) in scramble and shSrsf3 cell lines in the absence or presence (15-60 min) of PDGF-AA stimulation. Data are mean $\pm$ s.e.m. *, $P < 0.05$. Shaded shapes correspond to independent experiments. Summary statistics from biological replicates consisting of independent experiments (large shapes) are superimposed on top of data from all cells; $n = 20$ technical replicates across each of three biological replicates. (C-H) PDGFR α antibody signal (white or magenta) and Rab5 antibody signal (white or green) as assessed by immunofluorescence analysis of scramble and shSrsf3 cells in the absence or presence (15-60 min) of PDGF-AA stimulation. Nuclei were stained with DAPI (blue). White arrows denote colocalization. Scale bars: 20 μ m (main images), 3 μ m (insets). (I) Western blot (WB) analysis of whole-cell lysates (WCL) from scramble (left) and shSrsf3 (right) cell lines following a time course of PDGF-AA stimulation from 15 min to 4 h, with anti-phospho-(p)-Akt and anti-Akt antibodies. Line graphs depicting quantification of band intensities from $n = 3$ biological replicates as above. Data are mean $\pm$ s.e.m. *, $P < 0.05$; **, $P < 0.01$. (J) Model of experimental results in which PI3K/Akt-mediated PDGFR α signaling results in the nuclear translocation of Srsf3 and the subsequent AS of transcripts to decrease levels of proteins that promote PDGFR α trafficking out of early endosomes.

PDGFR α and Rab5 by 60 min of ligand treatment (0.186 ± 0.0102 PCC) (**Fig. 6B, D-D", F-F", H-H"** [↗](#)), indicating that Srsf3 activity downstream of PDGFR α signaling results in retention of PDGFR α in early endosomes.

Finally, we previously demonstrated that rapid internalization of PDGFR α homodimers following PDGF-AA ligand stimulation is critical for downstream AKT phosphorylation (Rogers et al., 2022 [↗](#)). As such, we examined phospho-Akt levels as a readout of PDGFR α activation in early endosomes. While PDGF-AA treatment for 15 min induced a robust phospho-Akt response in scramble cells (13.5 ± 7.39 relative induction, mean $\pm$ s.e.m.) this response was muted in shSrsf3 cells at the same timepoint (5.73 ± 1.91) (**Fig. 6I** [↗](#)). These findings further suggest that retention of PDGFR α in early endosomes leads to increases in downstream PI3K-mediated Akt signaling. Collectively, our data point to a feedback loop in which PI3K/Akt-mediated PDGFR α signaling results in the nuclear translocation of Srsf3 and the subsequent AS of transcripts to decrease levels of proteins that promote PDGFR α trafficking out of early endosomes (**Fig. 6J** [↗](#)).

Discussion

In this study, we confirmed our prior *in vivo* results in the mouse embryonic facial mesenchyme (Dennison et al., 2021 [↗](#)) that PDGFR α signaling primarily regulates gene expression via AS. PDGF-AA-dependent differential gene expression was minimal following one hour of ligand treatment and led to increased expression of immediate early genes *Klf10*, *Egf3* and *Egr1*, consistent with previous findings (Vasudevan et al., 2015 [↗](#)) and in line with the enriched GO terms of regulation of transcription and DNA binding. Alternatively, PDGF-AA-dependent alternatively-spliced transcripts were enriched for protein kinase activity, consistent with our prior AS findings upon disruption of PI3K-mediated PDGFR α signaling in the palatal shelf mesenchyme (Dennison et al., 2021 [↗](#)). Importantly, our results demonstrated a significant dependence on the RBP Srsf3 for AS downstream of PDGFR α activation. In fact, we found that 79% of Srsf3-dependent and 97% of PDGF-AA-dependent alternatively-spliced transcripts were responsive to both Srsf3 activity and PDGFR α signaling. As discussed above, Srsf3 is phosphorylated downstream of multiple stimuli (Bavelloni et al., 2014 [↗](#); Dennison et al., 2021 [↗](#); Fantauzzo & Soriano, 2014 [↗](#); Long et al., 2019 [↗](#); Zhou et al., 2012 [↗](#)), and it is likely that these additional inputs contributed to Srsf3-dependent AS that was non-responsive to PDGF-AA treatment. Further, our previous mass spectrometry screen identified 11 additional RBPs involved in AS that are phosphorylated by Akt downstream of PI3K-mediated PDGFR α signaling in primary MEPM cells (Fantauzzo & Soriano, 2014 [↗](#)), which may account for the small fraction of PDGF-AA-dependent AS that was non-responsive to Srsf3 knockdown. However, our RNA-seq results together with the phenotypic overlap of embryos with neural crest-specific ablation of *Srsf3* and mutant mouse models of *Pdgfra* and/or its ligands (Andrae et al., 2016 [↗](#); Dennison et al., 2021 [↗](#); Ding et al., 2004 [↗](#); Fantauzzo & Soriano, 2014 [↗](#); Fredriksson et al., 2012 [↗](#); Klinghoffer et al., 2002 [↗](#); Soriano, 1997 [↗](#)) make a compelling case for Srsf3 serving as a critical effector of PDGFR α signaling in the facial mesenchyme.

Here, we observed that Srsf3-dependent skipped exon events were enriched for transcripts with a + Δ PSI (exon skipped more often in shSrsf3 samples) in the absence of PDGF-AA stimulation, consistent with Srsf3 promoting exon inclusion. Alternatively, PDGF-AA ligand treatment led to an increase in the percentage of Srsf3-dependent transcripts with a - Δ PSI (exon included more often in shSrsf3 samples), suggesting that PDGFR α signaling causes decreased exon inclusion in the presence of Srsf3. Interestingly, a recent paper found that tethering Srsf3 downstream of the 5' splice site or upstream of the 3' splice site using MS2 stem loops did not lead to AS of a splicing reporter (Schmok et al., 2024 [↗](#)). However, the assay did not test tethering within the exon and used a single minigene sequence context, and thus had the potential to lead to false negative results (Schmok et al., 2024 [↗](#)). Whether phosphorylation of Srsf3 directly influences its binding to target RNAs or acts to modulate Srsf3 protein-protein interactions which then contribute to differential RNA binding remains to be determined, though findings from Schmok et al., 2024 [↗](#)

may argue for the latter mechanism. Studies identifying proteins that differentially interact with Srsf3 in response to PDGF-AA ligand stimulation are ongoing and will shed light on these mechanisms.

This study represents the first endogenous CLIP analysis of Srsf3 in the absence of protein tagging, and thus circumvents potential limitations with prior approaches in which assayed RBPs were overexpressed and/or fused to another protein. Our eCLIP analyses revealed several changes in Srsf3 transcript binding downstream of PDGF-AA stimulation, including increased Srsf3 binding to exons and loss of Srsf3 binding to canonical CA-rich motifs. A previous CLIP study using a stably integrated SRSF3-EGFP transgene in mouse P19 cells determined that SRSF3 binding was enriched in exons, particularly within 100 nucleotides of the 5' and, to a lesser extent, 3' splice sites (Ånkö et al., 2012 [↗](#)), consistent with our results. This same study identified a CA-rich canonical SRSF3 motif (Ånkö et al., 2012 [↗](#)). While such motifs were identified here in the absence of PDGF-AA treatment, they were lost upon ligand stimulation. Again, this shift could be due to loss of RNA binding owing to electrostatic repulsion and/or changes in ribonucleoprotein composition and will be the subject of future studies.

Our findings additionally pointed to novel properties of exons whose inclusion is dependent on Srsf3 in the absence of PDGFRα signaling. We demonstrated that these included exons had a higher GC content, a lower ratio of downstream intron to exon GC content and a decreased ratio of upstream and downstream intron to exon length. These findings are consistent with previous results demonstrating that included exons tend to have higher GC content than the flanking introns (Amit et al., 2012 [↗](#)). Of note, PDGF-AA ligand stimulation resulted in binding of Srsf3 to exons with a lower GC content, again suggesting that phosphorylation of the RBP downstream of this signaling axis promotes exon skipping.

By cross-referencing our RNA-seq and eCLIP datasets, we showed that 24% of the alternatively-spliced transcripts across our four treatment comparisons had an Srsf3 eCLIP peak. As Srsf3 also has functions in the cytoplasm, such as RNA trafficking, translation and degradation (Howard & Sanford, 2015 [↗](#)), the additional eCLIP peaks may reflect alternate roles for Srsf3 in RNA metabolism. Conversely, Srsf3-mediated AS may be delayed following transcript binding in the short timeframe of our experimental design. However, the extent of overlap that we observed is in line with previous studies correlating alternatively-spliced transcripts upon knockdown of an RBP with endogenous eCLIP results for that same RBP, including Rbfox2 (10%) (Moss et al., 2023 [↗](#)) and LUC7L2 (18-26%) (Jourdain et al., 2021 [↗](#)). The degree to which our RNA-seq and eCLIP datasets overlapped here points to the robustness and biological significance of our findings.

Our combined analyses demonstrated that Srsf3 binds and mediates the AS of transcripts that encode proteins that regulate PI3K signaling and early endosomal trafficking downstream of PDGFRα activation, including *Becn1*, *Wdr81* and *Acap3*. Homozygous missense mutations in *WDR81* cause Cerebellar ataxia, impaired intellectual development, and dysequilibrium syndrome 2 (OMIM 610185) in humans (Gulsuner et al., 2011 [↗](#)), with some patients exhibiting coarse facial features and strabismus (Garcias & Roth, 2007 [↗](#)), pointing to a critical role for this gene product in craniofacial development. Consistently, we found that related GO terms, such as phosphatidylinositol phosphate binding and endosome to lysosome transport, were significantly enriched among alternatively-spliced transcripts in murine embryonic facial mesenchyme upon loss of PI3K binding to PDGFRα and/or knockdown of *Srsf3* (Dennison et al., 2021 [↗](#)). These data further confirm that our iMEPM model system served as a powerful platform to uncover mechanisms that are utilized during craniofacial development *in vivo*.

Our subsequent *in vitro* validation studies showed that Srsf3-mediated PDGFRα signaling results in enlarged early endosomes, retention of the receptor in these early endosomes and increased downstream PI3K-mediated Akt signaling. Relatedly, we and others have linked spatial organization with the propagation of PDGFRα signaling, such that rapid internalization of the

receptors into early endosomes or autophagy of the receptors are required for maximal phosphorylation of AKT downstream of PDGFR α activation (Rogers et al., 2022 [↗](#); Simpson et al., 2024 [↗](#)). Together, these results indicate a feedback loop at play in the craniofacial mesenchyme in which stimulation of PDGFR α homodimer signaling leads to Srsf3-dependent AS of transcripts, a subsequent increase in the levels of proteins that maintain the receptor in early endosomes and a corresponding decrease in the levels of proteins that promote trafficking of the receptor to late endosomes for eventual degradation. These findings thus represent a novel mechanism by which PDGFR α activity is maintained and propagated within the cell. Whether similar mechanisms exist downstream of alternate RTKs or contribute to increases in the phosphorylation of effector proteins other than Akt remain to be determined.

Taken together, our findings significantly enhance our understanding of the molecular mechanisms by which Srsf3 activity is regulated downstream of growth factor stimulation. Interestingly, a recent study demonstrated that retention of another shuttling SR protein, Srsf1, exclusively in the nucleus resulted in widespread ciliary defects in mice (Haward et al., 2021 [↗](#)). This finding indicates that dissecting nuclear from cytoplasmic functions of SR proteins can provide powerful insight into the physiological relevance of each. Going forward, it will be critical to explore the *in vivo* consequences of abrogating Akt-mediated phosphorylation of Srsf3 and comparing the resulting phenotype to those of embryos with constitutive or conditional ablation of *Srsf3* in the neural crest lineage (Dennison et al., 2021 [↗](#); Jumaa et al., 1999 [↗](#)). These experiments are ongoing and should shed considerable light on the roles of RBP post-translational modifications during development.

Materials and Methods

Generation of scramble and Srsf3 shRNA iMEPM cell lines

Immortalized mouse embryonic palatal mesenchyme (iMEPM) cells were derived from a male *Cdkn2a*^{-/-} embryo as previously described (Fantauzzo & Soriano, 2017 [↗](#)). iMEPM cells were cultured in growth medium [Dulbecco's modified Eagle's medium (Gibco, Thermo Fisher Scientific, Waltham, MA, USA) supplemented with 50 U/mL penicillin (Gibco), 50 μ g/mL streptomycin (Gibco) and 2 mM L-glutamine (Gibco) containing 10% fetal bovine serum (FBS) (Hyclone Laboratories Inc., Logan, UT, USA)] and grown at 37°C in 5% carbon dioxide. iMEPM cells were tested for mycoplasma contamination using the MycoAlert Mycoplasma Detection Kit (Lonza Group Ltd, Basel, Switzerland). Packaged lentiviruses containing pLV[shRNA]-EGFP:T2A:Puro-U6>Scramble_shRNA (vectorID: VB010000-0009mxc) with sequence CCTAAGGTTAAGTCGCCCTCGCTCGAGCGAGGGCGACTTAACCTTAGG or pLV[shRNA]-EGFP:T2A:Puro-U6>mSrsf3[shRNA#1] (vectorID: VB900060-7699yyh) with sequence GAATGATAAAGCGGTGTTTACTCGAGTAAACACCGCTTTATCATTC were purchased from VectorBuilder (Chicago, IL, USA). Medium containing lentivirus for a multiplicity of infection of 10 for 200,000 cells with the addition of 10 μ g/mL polybrene was added to iMEPM cells for 16 h, and cells were subsequently grown in the presence of 4 μ g/mL puromycin for 10 days. Cells with the highest GFP expression (top 20%) were isolated on a Moflo XDP 100 cell sorter (Beckman Coulter Inc., Brea, CA, USA) and expanded. Srsf3 expression in scramble and *Srsf3* shRNA cell lines was confirmed by western blotting. Once the stable cell lines were established, they were split at a ratio of 1:4 for maintenance. Scramble and *Srsf3* shRNA cells were used for experiments at passages 9-20.

Immunoprecipitation and western blotting

To induce PDGFR α signaling, cells at ~70% confluence were serum starved for 24 h in starvation medium [Dulbecco's modified Eagle's medium (Gibco) supplemented with 50 U/mL penicillin (Gibco), 50 μ g/mL streptomycin (Gibco) and 2 mM L-glutamine (Gibco) containing 0.1% FBS (Hyclone Laboratories Inc.)] and stimulated with 10 ng/mL rat PDGF-AA ligand (R&D Systems,

Minneapolis, MN, USA) diluted from a 1.5 µg/mL working solution in 40 nM HCl containing 0.1% BSA for the indicated length of time. When applicable, UV-crosslinking was performed at 254 nm and 400 mJ/cm² using a Vari-X-Link system (UVO3 Ltd, Cambridgeshire, UK). For immunoprecipitation of Srsf3, cells were resuspended in ice-cold CLIP lysis buffer [50 mM Tris-HCl pH 7.4, 100 mM NaCl, 1% NP-40, 0.1% SDS, 0.5% sodium deoxycholate, 1x complete Mini protease inhibitor cocktail (Roche, MilliporeSigma, Burlington, MA, USA), 1 mM PMSF, 10 mM NaF, 1 mM Na₃VO₄, 25 mM β-glycerophosphate]. Cleared lysates were collected by centrifugation at 18,000 *g* for 20 min at 4°C. Anti-Srsf3 antibody (10 µg/sample) (ab73891, Abcam, Waltham, MA, USA) was added to protein A Dynabeads (125 µL/sample) (Thermo Fisher Scientific) washed twice in ice-cold CLIP lysis buffer and incubated for 45 min at room temperature. Cells lysates were incubated with antibody-conjugated Dynabeads or Dynabeads M-280 sheep anti-rabbit IgG (Thermo Fisher Scientific) washed twice in ice-cold CLIP lysis buffer overnight at 4°C. The following day, Dynabeads were washed twice each with ice-cold high salt wash buffer [50 mM Tris-HCl pH 7.4, 1M NaCl, 1 mM EDTA, 1% NP-40, 0.1% SDS, 0.5% sodium deoxycholate] followed by ice-cold wash buffer [20 mM Tris-HCl pH 7.4, 10 mM MgCl₂, 0.2% Tween-20]. The precipitated proteins were eluted with 1x NuPAGE LDS buffer (Thermo Fisher Scientific) containing 100 mM dithiothreitol, heated for 10 minutes at 70°C, and separated by SDS-PAGE. For western blotting of whole-cell lysates, protein lysates were generated by resuspending cells in ice-cold NP-40 lysis buffer (20 mM Tris-HCl pH 8, 150 mM NaCl, 10% glycerol, 1% Nonidet P-40, 2 mM EDTA, 1x complete Mini protease inhibitor cocktail (Roche), 1 mM PMSM, 10 mM NaF, 1 mM Na₃VO₄, 25 mM β-glycerophosphate) and collecting cleared lysates by centrifugation at 13,400 *g* at 4°C for 20 min. Laemmli buffer containing 10% β-mercaptoethanol was added to the lysates, which were heated for 5 min at 100°C. Proteins were subsequently separated by SDS-PAGE. Western blot analysis was performed according to standard protocols using horseradish peroxidase-conjugated secondary antibodies. Blots were imaged using a ChemiDoc XRS+ (Bio-Rad Laboratories, Inc., Hercules, CA, USA) or a ChemiDoc (Bio-Rad Laboratories, Inc.). The following primary antibodies were used for western blotting: Srsf3 (1:1,000, ab73891, Abcam), Gapdh (1:50,000, 60004, Proteintech Group, Inc., Rosemont, IL, USA), phospho-Akt (Ser473) (1:1,000, 9271, Cell Signaling Technology, Inc., Danvers, MA, USA), Akt (1:1,000, 9272, Cell Signaling Technology, Inc.), horseradish peroxidase-conjugated goat anti-mouse IgG (1:20,000, 115035003, Jackson ImmunoResearch Inc., West Grove, PA, USA), horseradish peroxidase-conjugated goat anti-rabbit IgG (1:20,000, 111035003, Jackson ImmunoResearch Inc.). Quantifications of signal intensity were performed with ImageJ software (version 1.53t, National Institutes of Health, Bethesda, MD, USA). Relative phospho-Akt levels were determined by normalizing to total Akt levels. When applicable, statistical analyses were performed with Prism 10 (GraphPad Software Inc., San Diego, CA, USA) using a two-tailed, ratio paired *t*-test within each cell line and a two-tailed, unpaired *t*-test with Welch's correction between each cell line. Immunoprecipitation and western blotting experiments were performed across three independent experiments.

RNA-sequencing and related bioinformatics analyses

8 x 10⁵ cells obtained from each of three independent biological replicates per treatment were frozen on liquid nitrogen and stored at -80°C. Following thawing, total RNA was simultaneously isolated from all samples using the RNeasy Mini Kit (Qiagen, Inc., Germantown, MD, USA) according to the manufacturer's instructions. RNA was forwarded to the University of Colorado Cancer Center Genomics Shared Resource for quality control, library preparation, and sequencing. RNA purity, quantity and integrity were assessed with a NanoDrop (Thermo Fisher Scientific) and a 4200 TapeStation System (Agilent Technologies, Inc., Santa Clara, CA, USA) prior to library preparation. Total RNA (200 ng) was used for input into the Universal Plus mRNA-Seq kit with NuQuant (Tecan Group Ltd., Männedorf, Switzerland). Dual index, stranded libraries were prepared and sequenced on a NovaSeq 6000 Sequencing System (Illumina, San Diego, CA, USA) to an average depth of ~54 million read pairs (2×150 bp reads).

Raw sequencing reads were de-multiplexed using bcl2fastq (Illumina). Trimming, filtering and adapter contamination removal was performed using BBduk (from the BBmap v35.85 tool suite) (Bushnell, 2015 [DOI](#)). For differential expression analysis, transcript abundance was quantified using Salmon (v1.4.0) (Patro et al., 2017 [DOI](#)) and a decoy-aware transcriptome index prepared using GENCODE (Frankish et al., 2019 [DOI](#)) GRCm39 M26. Gene level summaries were calculated using tximport (Soneson et al., 2016 [DOI](#)) in R and differential expression was measured using DESeq2 (v1.32.0) (Love et al., 2014 [DOI](#)). Significant changes in gene-level expression are reported for cases with adjusted $P \leq 0.05$ and fold change $|FC| \geq 2$. Spearman correlation was computed between conditions for differentially-expressed genes. For alternative splicing analysis, raw FASTQ were trimmed to a uniform length of 125 bp. Reads were aligned to the mouse genome (GRCm39 Gencode M26) using STAR (v2.7.9a) (Dobin et al., 2013 [DOI](#)). Additional parameters for STAR: --outFilterType BySJout --outFilterMismatchNmax 10 --outFilterMismatchNoverLmax 0.04 --alignEndsType EndToEnd --runThreadN 16 --alignSJDBoverhangMin 4 --alignIntronMax 300000 --alignSJoverhangMin 8 --alignIntronMin 20. All splice junctions detected in at least 1 read from the first pass alignment were used in a second pass alignment, per software documentation. Alternative splicing events were detected using rMATS (v4.0.2, default parameters plus ‘--cstat 0.0001’) (Shen et al. 2014 [DOI](#)). Reads mapping to the splice junction as well as those mapping to the exon body were used in downstream analyses. Detected events were compared between treatment groups and considered significant with false discovery rate (FDR) ≤ 0.05 , a difference in percent spliced in ($|\Delta PSI|$) ≥ 0.05 and event detection in at least 10 reads in either condition. Raw read pairs, trimmed read pairs for Salmon input, Salmon mapping rate per sample, trimmed read pairs (125 bp) for STAR input and STAR unique mapping rate can be found in Table S1. Gene ontology analysis was performed with various libraries from the Enrichr gene list enrichment analysis tool (Chen et al., 2013 [DOI](#); Kuleshov et al., 2016 [DOI](#)) and terms with $P < 0.05$ were considered significant.

qPCR

Total RNA was isolated using the RNeasy mini kit (Qiagen, Germantown, MD, USA) according to the manufacturer’s instructions. First-strand cDNA was synthesized using a ratio of 2:1 random primers:oligo (dT) primer and SuperScript II RT (Invitrogen, Thermo Fisher Scientific) according to the manufacturer’s instructions. All reactions were performed with 1× ThermoPol buffer [0.02 M Tris (pH 8.8), 0.01 M KCl, 0.01 M $(\text{NH}_4)_2\text{SO}_4$, 2 mM MgSO_4 and 0.1% Triton X-100], 200 μM dNTPs, 200 nM primers (Integrated DNA Technologies, Inc., Coralville, IA, USA), 0.6 U Taq polymerase and 1 μg cDNA in a 25 μL reaction volume. The primers used can be found in Table S16. The following PCR protocol was used for *Arhgap12*: step1, 3 min at 94°C; step 2, 30 s at 94°C; step 3, 30 s at 47°C; step 4, 30 s at 72°C; repeat steps 2-4 for 34 cycles; and step 5, 5 min at 72°C. The following PCR protocol was used for *Cep55*: step1, 3 min at 94°C; step 2, 30 s at 94°C; step 3, 30 s at 48°C; step 4, 30 s at 72°C; repeat steps 2-4 for 34 cycles; and step 5, 5 min at 72°C. Two-thirds of total PCR products were electrophoresed on a 2% agarose/TBE gel containing ethidium bromide and photographed on an Aplegen Omega Fluor Gel Documentation System (Aplegen Inc., Pleasanton, CA, USA). Quantifications of band intensity were performed with ImageJ software (version 1.53t, National Institutes of Health). The PSI was calculated independently for each sample as the percentage of the larger isoform divided by the total abundance of all isoforms within the given gel lane. Statistical analyses were performed with Prism 10 (GraphPad Software) using a two-tailed, unpaired *t*-test with Welch’s correction. qPCR reactions were performed using three biological replicates.

Enhanced UV-crosslinking and immunoprecipitation and related bioinformatics analyses

Experiments were performed as previously described in biological duplicates (Van Nostrand et al., 2016 [DOI](#), 2017). Briefly, 2 million cells per treatment were serum starved and treated with 10 ng/mL PDGF-AA as described above. Cells were subsequently UV-crosslinked at 254 nm and 400 mJ/cm², scraped in 1x phosphate buffered saline (PBS) and transferred to 1.5 mL Eppendorf tubes, at which point excess PBS was removed and cells were frozen on liquid nitrogen and stored at -80°C.

Following thawing, cells were lysed in ice-cold CLIP lysis buffer, sonicated by BioRuptor (Diagenode, Denville, NJ, USA) and treated with RNase I (Thermo Fisher Scientific). 2% of lysates were set aside as size-matched input samples. Srsf3-RNA complexes were immunoprecipitated with anti-Srsf3 antibody (10 µg per sample) (ab73891, Abcam) conjugated to protein A Dynabeads (Thermo Fisher Scientific). IP samples were washed and dephosphorylated with FastAP (New England Biolabs, Ipswich, MA, USA) and T4 PNK (New England Biolabs). IP samples underwent on-bead ligation of barcoded RNA adapters (/5phos/rArGrArUrCrGrArArGrArGrUrCrGrUrG/3SpC3/) to the 3' end using T4 RNA ligase (New England Biolabs). Following elution, protein-RNA complexes were run on 4-12% Bis-Tris 1.5 mm gels (Thermo Fisher Scientific) and transferred onto nitrocellulose membranes. The 20-75 kDa region was excised and digested with proteinase K (New England Biolabs). RNA was isolated with acid phenol/chloroform/isoamyl alcohol (pH 6.5) (Thermo Fisher Scientific), reverse transcribed with Superscript III (Thermo Fisher Scientific) and treated with ExoSAP-IT (Affymetrix, Thermo Fisher Scientific) to remove excess primers and unincorporated nucleotides. Samples underwent 3' ligation of barcoded DNA adapters (/5Phos/NNNNNNNNNAGATCGGAAGAGCACAGTCTG/3SpC3/), clean-up with Dynabeads MyOne Silane (Thermo Fisher Scientific) and qPCR to determine the appropriate number of PCR cycles. Libraries were then amplified with Q5 PCR mix (New England Biolabs) for a total of 16-25 cycles. Libraries were forwarded to the University of Colorado Cancer Center Genomics Shared Resource for quality control and sequencing. Sample integrity was assessed with a D1000 ScreenTape System (Agilent Technologies, Inc.) and sequenced on a NovaSeq 6000 Sequencing System (Illumina) to an average depth of ~20 million read pairs (2×150 bp reads).

Raw sequencing reads were de-multiplexed using bcl2fastq (Illumina). Adapters were trimmed using cutadapt (v.1.18) (Martin, 2011 [↗](#)). Trimmed reads were quality filtered and collapsed using a combination of FASTX-Toolkit (v.0.0.14) (http://hannonlab.cshl.edu/fastx_toolkit [↗](#)), seqtk (v.1.3-r106) (<https://github.com/lh3/seqtk> [↗](#)) and custom scripts. After collapsing the reads, unique molecular identifiers were removed using seqtk. STAR index for repetitive elements was created using repetitive sequences from msRepDB (Liao et al., 2022 [↗](#)). Reads ≥ 18 nt were mapped to the repetitive elements using STAR (v.2.7.9a) (Dobin et al. 2013 [↗](#)). Reads unmapped to the repetitive elements were mapped to the mouse genome (GRCm39 Gencode M26) using STAR (v.2.7.9a) with parameters alignEndsType: EndtoEnd and outFilterMismatchNoverReadLmax: 0.04. Peaks were called using omniCLIP (v.0.20) (Drewe-Boss et al., 2018 [↗](#)) with the foreground penalty (--fg_pen) parameter set to 5. Peaks were annotated and motif analyses performed using RCAS (v.1.19.0) (Uyar et al., 2017 [↗](#)) and custom R script. For visualization purposes, bigWig files were created from bam files using deepTools (v.3.5.5) (Ramírez et al., 2016 [↗](#)). Peaks were visualized in Integrative Genomics Viewer (v.2.13.0) (Robinson et al., 2011 [↗](#)). Intron and exon features were calculated using Matt (v.1.3.1) (Gohr & Irimia, 2019 [↗](#)), and statistical analyses were performed using a Mann-Whitney U test. For overlap of eCLIP peaks and alternative splicing events, peak coordinates were taken from omniCLIP bed files and alternative splicing coordinates were taken from rMATS output. Overlapping coordinates from alternative splicing events were defined following the rMAPS default values (Park et al., 2016 [↗](#)). Overlap was calculated using valr (v.0.6.4) (Riemondy et al., 2017 [↗](#)) and custom R scripts. Raw read pairs, trimmed read pairs, collapsed reads, reads after removing repetitive elements, mapped reads, peaks and annotated peaks can be found in Table S7. Gene ontology analysis was performed with various libraries from the Enrichr gene list enrichment analysis tool (Chen et al., 2013 [↗](#); Kuleshov et al., 2016 [↗](#)) and terms with $P < 0.05$ were considered significant.

Immunofluorescence analysis

Cells were seeded onto glass coverslips at ~40% confluency per 24-well plate well in iMEPM growth medium. After 24 h, cells were serum starved and treated with 10 ng/mL PDGF-AA as described above. Cells were fixed in 4% paraformaldehyde (PFA) in PBS with 0.1% Triton X-100 for 10 min and washed in PBS. Cells were blocked for 1 h in 5% normal donkey serum (Jackson ImmunoResearch Inc.) in PBS and incubated overnight at 4°C in primary antibody diluted in 1%

normal donkey serum in PBS. After washing in PBS, cells were incubated in Alexa Fluor 488-conjugated donkey anti-rabbit secondary antibody (1:1,000; A21206; Invitrogen) or Alexa Fluor 546-conjugated donkey anti-mouse secondary antibody (1:1,000; A10036; Invitrogen) diluted in 1% normal donkey serum in PBS with 2 $\mu\text{g}/\text{ml}$ DAPI (Sigma-Aldrich, St. Louis, MO, USA) for 1 h. Cells were mounted in VECTASHIELD HardSet Antifade Mounting Medium (Vector Laboratories, Inc., Burlingame, CA, USA) and photographed using an Axiocam 506 mono digital camera (Carl Zeiss Microscopy LLC, White Plains, NY, USA) fitted onto an Axio Observer 7 fluorescence microscope (Carl Zeiss Microscopy LLC) with the 63x oil objective with a numerical aperture of 1.4 at room temperature. The following antibodies were used for immunofluorescence analysis: Rab5 (1:200, C8B1, 3547, Cell Signaling Technology Inc.), PDGFR α (1:20, AF1062, R&D Systems). For assessment of Rab5 puncta size and colocalization experiments, three independent trials, or biological replicates, were performed. For each biological replicate, 20 technical replicates consisting of individual cells were imaged with Z-stacks (0.24 μm between Z-stacks with a range of 1–6 Z-stacks) per timepoint. Images were deconvoluted using ZEN Blue software (Carl Zeiss Microscopy LLC) using the ‘Better, fast (Regularized Inverse Filter)’ setting. Extended depth of focus was applied to Z-stacks using ZEN Blue software (Carl Zeiss Microscopy LLC) to generate images with the maximum depth of field. For assessment of Rab5 puncta size, images were converted to 8-bit using Fiji software (version 2.14.0/1.54f). Images were subsequently converted to a mask and watershed separation was applied. A region of interest (ROI) was drawn around each Rab5-positive cell and particles were analyzed per cell using the “analyze particles” function. For colocalization measurements, an ROI was drawn around each PDGFR α -positive cell in the corresponding Cy3 (marker) channel using Fiji. For each image with a given ROI, the Cy3 channel and the EGFP channel were converted to 8-bit images. Colocalization was measured using the Colocalization Threshold function, where the rcoloc value [Pearson’s correlation coefficient (PCC)] was used in statistical analysis. Statistical analyses were performed on the average values from each biological replicate with Prism 10 (GraphPad Software Inc.) using a two-way ANOVA followed by uncorrected Fisher’s LSD test.

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Competing Interests

The authors declare no competing or financial interests.

Author contributions

Conceptualization: TEF, MS, NM, KAF; Methodology: TEF, MS, NM, KAF; Formal analysis: TEF, MS, EDL, KAF; Investigation: TEF, MS, EDL; Writing – Original Draft: TEF, KAF; Writing – Review & Editing: MS, EDL, NM; Visualization: TEF, MS, KAF; Supervision: NM, KAF; Project administration: KAF; Funding acquisition: TEF, KAF.

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Data Availability

The eCLIP and RNA-sequencing datasets generated during this study have been deposited in GEO under SuperSeries accession number GSE263170. Custom analysis scripts will be provided by Dr. Larson through GitHub.

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Reviewer #1 (Public Review):

In their manuscript "PDGFRRa signaling regulates Srsf3 transcript binding to affect PI3K signaling and endosomal trafficking" Forman and colleagues use iMEPM cells to characterize the effects of PDGF signaling on alternative splicing. They first perform RNA-seq using a one-hour stimulation with Pdgf-AA in control and Srsf3 knockdown cells. While Srsf3 manipulation results in a sizeable number of DE genes, PDGF does not. They then turn to examine alternative splicing, due to findings from this lab. They find that both PDGF and Srsf3 contribute much more to splicing than transcription. They find that the vast majority of PDGF-mediated alternative splicing depends upon Srsf3 activity and that skipped exons are the most common events with PDGF stimulation typically promoting exon skipping in the presence of Srsf3. They used eCLIP to identify RNA regions bound to Srsf3. Under both PDGF conditions, the majority of peaks were in exons with +PDGF having a substantially greater number of these peaks. Interestingly, they find differential enrichment of sequence motifs and GC content in stimulated versus unstimulated cells. They examine 2 transcripts encoding PI3K pathway (enriched in their GO analysis) members: Becn1 and Wdr81. They then go on to examine PDGFRRa and Rab5, an endosomal marker, colocalization. They propose a model in which Srsf3 functions downstream of PDGFRRa signaling to, in part, regulate PDGFRRa trafficking to the endosome. The findings are novel and shed light on the mechanisms of PDGF signaling and will be broadly of interest. This lab previously identified the importance of PDGF signaling on alternative splicing. The combination of RNA-seq and eCLIP is an exceptional way to comprehensively analyze this effect. The results will be of great utility to those studying PDGF signaling or neural crest biology. There are some concerns that should be considered, however.

(1) It took some time to make sense of the number of DE genes across the results section and Figure 1. The authors give the total number of DE genes across Srsf3 control and loss conditions as 1,629 with 1,042 of them overlapping across Pdgf treatment. If the authors

would add verbiage to the point that this leaves 1,108 unique genes in the dataset, then the numbers in Figure 1D would instantly make sense. The same applies to PDGF in Figure 1F and the Venn diagrams in Figure 2.

(2) The percentage of skipped exons in the + Δ PSI on the righthand side of Figure 2F is not readable.

(3) It would be useful to have more information regarding the motif enrichment in Figure 3. What is the extent of enrichment? The authors should also provide a more complete list of enriched motifs, perhaps as a supplement.

(4) It is unclear what subset of transcripts represent the "overlapping datasets" on lines 280-315. The authors state that there are 149 unique overlapping transcripts, but the Venn diagram shows 270. Also, it seems that the most interesting transcripts are the 233 that show alternative splicing and are bound by Srsf3. Would the results shown in Figure 5 change if the authors focused on these transcripts?

(5) In general, there is little validation of the sequencing results, performing qPCR on Arhgap12 and Cep55. The authors should additionally validate the PI3K pathway members that they analyze. Related, is Becn1 expression downregulated in the absence of Srsf3, as would be predicted if it is undergoing NMD?

(6) What is the alternative splicing event for Acap3?

(7) The insets in Figure 6 C"-H" are useful but difficult to see due to their small size. Perhaps these could be made as their own figure panels.

(8) In Figure 6A, it is not clear which groups have statistically significant differences. A clearer visualization system should be used.

(9) Similarly in Figure 6B, is 15 vs 60 minutes in the shSrsf3 group the only significant difference? Is there a difference between scramble and shSrsf3 at 15 minutes? Is there a difference between 0 and 15 minutes for either group?

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Reviewer #2 (Public Review):

Summary:

This manuscript builds upon the work of a previous study published by the group (Dennison, 2021) to further elucidate the coregulatory axis of Srsf3 and PDGFRa on craniofacial development. The authors in this study investigated the molecular mechanisms by which PDGFRa signaling activates the RNA-binding protein Srsf3 to regulate alternative splicing (AS) and gene expression (GE) necessary for craniofacial development. PDGFRa signaling-mediated Srsf3 phosphorylation drives its translocation into the nucleus and affects binding affinity to different proteins and RNA, but the exact molecular mechanisms were not known. The authors performed RNA sequencing on immortalized mouse embryonic mesenchyme (MEPM) cells treated with shRNA targeting 3' UTR of Srsf3 or scramble shRNA (to probe AS and DE events that are Srsf3 dependent) and with and without PDGF-AA ligand treatment (to probe AS and DE events that are PDGFRa signaling dependent). They found that PDGFRa signaling has more effect on AS than on DE. A matching eCLIP-seq experiment was performed to investigate how Srsf3 binding sites change with and without PDGFRa signaling.

Strengths:

(1) The work builds well upon the previous data and the authors employ a variety of appropriate techniques to answer their research questions.

(2) The authors show that Srsf3 binding pattern within the transcript as well as binding motifs change significantly upon PDGFRa signaling, providing a mechanistic explanation for the significant changes in AS.

(3) By combining RNA-seq and eCLIP datasets together, the authors identified a list of genes that are directly bound by Srsf3 and undergo changes in GE and/or AS. Two examples are Becn1 and Wdr81, which are involved in early endosomal trafficking.

Weaknesses:

(1) The authors identify two genes whose AS are directly regulated by Srsf3 and involved in endosomal trafficking; however, they do not validate the differential AS results and whether changes in these genes can affect endosomal trafficking. In Figure 6, they show that PDGFRa signaling is involved in endosome size and Rab5 colocalization, but do not show how Srsf3 and the two genes are involved.

(2) The proposed model does not account for other proteins mediating the activation of Srsf3 after Akt phosphorylation. How do we know this is a direct effect (and not a secondary or tertiary effect)?

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